

Damian Lapena Vidal

damian.lapenavidal@gmail.com | +1 (401) 441-2830 | <https://damianlapenavidal.github.io/> | <https://www.linkedin.com/in/damianlapevi/> | Kingston, RI

EDUCATION

University of Rhode Island (URI) Kingston, RI	Expected May 2027
Bachelor of Science in Computer Engineering	GPA: 4.0/4.0
• Scholarships: URI Presidential, Subramaniam Family, C & W Eckman, Dr. & Mrs. E. Eddy, Barbara A. Woods Memorial • Courses: Machine Learning I (Python/Scikit-Learn), Data Structures & Algorithms (C++), Electronics I (LTSpice), Linear Circuit Theory (MATLAB), Digital Circuit Design (NI Multisim), Digital Design with FPGAs (VHDL), Computer Architecture • Languages: Spanish (Native), English (Fluent/Bilingual), German (Basic)	

ENGINEERING EXPERIENCE

Student Technician URI IT Services	January 2026 – Present
• Using Q-SYS to monitor classroom tech across campus, making sure all hardware and digital interfaces are running smoothly • Acting as the first point of contact for professors facing technical issues, providing quick on-site fixes or remote solutions • Handling hands-on hardware work, from swapping out projectors to replacing cable infrastructure • Work closely with a team of six to coordinate schedules and managing the ticketing system for all repairs	
Quantum Computing Program QuLearnLabs	May 2025 – July 2025
• Learned principles of quantum mechanics relevant to computing, post-quantum cryptography, and quantum-safe protocols • Designed and simulated quantum circuits using IBM Qiskit SDK • Built a capstone project implementing Grover's Algorithm using 4 qubits	
URI Undergraduate Research Assistant URI Integrated Circuits Design Lab	September 2024 – May 2025
• Assisted a team of three in PCB design using Altium Designer, incorporating transistors for cryogenic and radiation testing • Performed manual soldering for prototype assemblies and ran standard condition simulations using LTSpice • Conducted electrical characterization by measuring I-V curves using power supply and multimeter under standard conditions • Developed a Python-based parameter optimization framework using PyTorch, integrating NGSpice for circuit simulation, finite-difference gradients for optimization, Git/GitHub for version control, and Weights & Biases for hyperparameter evaluation	

PROJECTS

Campus AI Assistant (RAG System) URI College of Engineering	November 2025 – Present
• Developing a Retrieval-Augmented Generation (RAG) AI assistant to automate responses to high-volume student inquiries • Using Python, LangChain, and OpenAI APIs to generate context-aware responses from institutional data • Implementing real-time web scraping and preprocessing pipelines using Firecrawl and BeautifulSoup4 to collect university data • Containerizing the system/application with Docker to support reproducible development and scalable deployment • Collaborating in a 4-person team to design the retrieval pipeline, data ingestion workflows, and evaluation of AI responses	
Non-Linear Circuit Analysis & Harmonic Distortion Electronics I Lab	October 2025
• Designed and tested a BJT amplifier to characterize non-linear signal distortion and identify transition points between linear, saturation, and cutoff regions • Utilized oscilloscope FFT functions to perform frequency domain analysis, isolating and measuring 2nd & 3rd order harmonics • Validated breadboard results against LTSpice transient and DC operating point simulations to ensure hardware accuracy	
DigiSign HackHarvard '25 (https://digisign-6e9.pages.dev/)	October 2025
• Innovated a JavaScript-based e-signature capture tool that tracks a user's finger via webcam for secure payment transactions • Built an interactive frontend using React, focusing on delivering smooth UX and an intuitive, user-friendly UI • Developed backend services in Node.js, utilizing Google Gemini API for real-time signature processing and data storage • Integrated Stripe API for safe wallet transactions, real-time payment verification, and automated billing management • Handled deployment with Cloudflare and managed dependencies via Node Package Manager for efficient, scalable delivery • Collaborated in a high-pressure team environment, leveraging strong communication, adaptability, and time management to deliver a functional MVP within the 36 hours hackathon deadline	

LEADERSHIP EXPERIENCE

URI Quantum Technologies Club Founder & President	February 2026 – Present
• Founding the university's first student organization dedicated to Quantum Tech, focusing on the technology's future impact • Leading a student executive board to handle the club's operations and help grow a meaningful quantum community • Connecting with experts from industry and academia to invite guest speakers and give students a look at real-world careers	
QuLearnLabs Ambassador	February 2026 – Present
• Advancing QuLearnLab's mission to expand quantum literacy	
Hack@URI Organizer	June 2025 – Present
• Helping in the planning, coordination, and behind-the-scenes work that brings a major tech event to URI • Collaborating closely with other teams to manage expenses, organize reimbursements, and support sponsor follow-ups • Attending leading hackathons such as HackHarvard to gather insights and implement innovative ideas for Hack@URI	

CAMPUS INVOLVEMENT

Tau Beta Pi National Engineering Honor Society Member	December 2025 – Present
URI Investment Club Member	September 2025 – Present
URI Pickleball Club Member	September 2024 – May 2025
Society of Hispanic Professional Engineers Member	March 2024 – Present